

Like the ass reaching for the sun at the apex of Dürer's woodcut, we tend to be unwitting to the consequences of mean reversion. Behavioral psychologist Daniel Kahneman, winner of the 2002 Nobel Prize in economics, recounts in his autobiography that his first observation of this unawareness was "the most satisfying Eureka experience" of his career.²¹ While attempting to lecture a group of Israeli air force flight instructors that praise is more effective than punishment for promoting learning, Kahneman was interrupted by one of the most experienced instructors in the audience. He began by conceding that positive reinforcement might work in the wider community, but denied that it was ideal for flight cadets. He said, "On many occasions I have praised flight cadets for clean execution of some aerobatic maneuver, and in general when they try it again, they do worse. On the other hand, I have often screamed at cadets for bad execution, and in general they do better the next time. So please don't tell us that reinforcement works, and punishment does not, because the opposite is the case." Kahneman says that he then understood an important truth about the world: Because we tend to reward others when they do well, and punish them when they do badly, and because there is regression to the mean, it is part of the human condition that we are statistically punished for rewarding others and rewarded for punishing them. He immediately arranged a demonstration in which each participant tossed two coins at a target behind his back, without any feedback. They measured the distances from the target and could see that those who had done best the first time had mostly deteriorated on their second try, and vice versa. Kahneman knew that this demonstration would not undo the effects of "lifelong exposure to a perverse contingency," but it set him on a path that would eventually lead to the Nobel Memorial Prize in economics. The evidence is that our attitude to mean reversion in stock prices is informed by the same perverse contingency.

Behavioral finance researchers Joseph Lakonishok, Andrei Shleifer, and Robert Vishny conducted a landmark study in 1994 into the reasons why value strategies beat the market. Lakonishok et al. concluded that they do so because value strategies are contrarian to the "naïve" strategies followed by other investors who, like the asses clinging to the wheel in Dürer's woodcut, fail to fully appreciate the implications of mean reversion.²² These naive strategies might range from extrapolating past earnings growth too far into the future; to assuming a trend in stock prices; to overreacting to good or bad news; or to simply equating a good investment with a well-run company irrespective of price. Some investors get overly excited about stocks that have done well in the past and bid them up, so that these so-called glamour stocks become overvalued. Similarly, they overreact to stocks that have performed poorly, oversell them, and these out-of-favor so-called value stocks become undervalued. As De Bondt and Thaler demonstrated,